

Metro Ticket Booking System

User Acceptance Testing (UAT) Strategy Specification (Project Planning Phase)

PLATFORM TARGET
ServiceNow App Engine

PROJECT PHASE
Project Planning Phase

DOCUMENT VERSION
v1.0 Final

1. Scope & Objective of UAT

The goal of User Acceptance Testing (UAT) is to validate that the ServiceNow Metro Ticket Booking system meets business requirements and is ready for production deployment from an end-user perspective.

In-Scope User Roles & Personas

- **Commuter / Passenger:** End users booking single-journey or pass tickets via Service Portal or Mobile App (Now Mobile).
- **Station Master / Gate Operator:** Station staff overseeing turnstiles, verifying ticket issues, and scanning backup tickets.
- **Transit Admin / Finance Team:** Back-office staff managing station fare charts, refund approvals, and daily revenue audit reports.

2. Key UAT Scenarios & Business Workflows

ID	Test Scenario / Persona	Acceptance Criteria	Priority
UAT-01	Single Journey Ticket Booking (Passenger)	Passenger selects origin/destination stations, calculates fare, pays via gateway, and instantly receives a valid QR code ticket.	CRITICAL
UAT-02	Turnstile Gate Scan & Entry (Passenger / Gate Operator)	Commuter scans QR code at entry gate. System validates active status in <150ms, opens gate, and updates status to In-Transit.	CRITICAL
UAT-03	Exit Gate Validation & Closure (Passenger / Gate Operator)	Commuter scans QR code at destination. Gate opens, ticket status updates to Used / Completed, and QR code is invalidated for reuse.	CRITICAL
UAT-04	Over-travel / Extended Journey (Passenger / Station Master)	If passenger exits at a farther station than booked, gate remains closed and prompts passenger to pay excess fare at Customer Care desk.	HIGH
UAT-05	Ticket Cancellation & Refund (Passenger)	Passenger cancels ticket before expiry/commence time; system calculates eligible refund per rules and triggers payment gateway refund.	MEDIUM
UAT-06	Fare Matrix Modification (Transit Admin)	Admin updates fare between Station A and Station B in ServiceNow Decision Tables; new bookings instantly reflect updated pricing.	HIGH

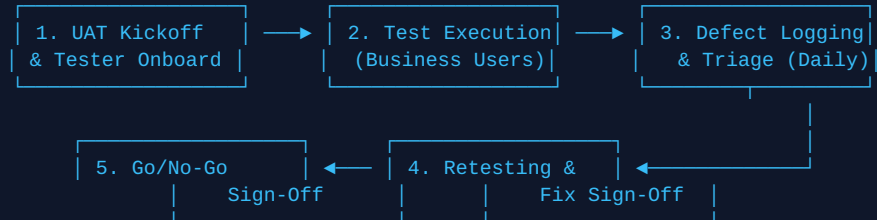
3. UAT Environment & Data Requirements

- **Dedicated Instance:** Conduct UAT on a dedicated **Test or Staging Instance** (cloned from production baseline).
- **Mock Hardware / Gate Simulators:** Turnstile gates or gate REST API mock utilities connected to the test instance.
- **Test Payment Gateway:** Payment sandbox mode (Stripe / Razorpay / Mock Gateway) enabled with test payment credentials.
- **Pre-seeded Master Data:**
 - Active stations, lines/routes, and zone-based fare structures.
 - Test user accounts with appropriate roles (sn_metro.commuter, sn_metro.station_operator, sn_metro.admin).

4. Acceptance Criteria & Go/No-Go Metrics

- [✓] **Zero Blockers / Critical Defects:** 100% of Critical and High priority UAT test cases must pass without open defects.
- [✓] **Usability Threshold:** Commuter ticket purchasing workflow must take no more than 3 steps / 60 seconds on mobile/portal.
- [✓] **Business Approval:** Formal sign-off from key stakeholders (Operations Lead, Customer Service Manager, Finance Lead).

5. UAT Execution Process & Governance



- Kickoff:** Onboard business testers with brief training and step-by-step UAT scripts.
- Execution & Logging:** Business users execute scenarios and log issues directly in ServiceNow.
- Defect Triage:** Daily 15-minute triage meeting during UAT execution to classify defects (Blocker, Major, Minor).
- Retesting & Fix Verification:** Development team resolves issues and business users retest for sign-off.
- Go/No-Go Sign-Off:** Final evaluation against acceptance metrics to approve deployment.